

IronMind: Scaling Humanoid Dexterous Manipulation via Camera-Space Ego-Centric Pretraining

World Model Team, XPENG Robotics

Project: <https://xpeng-robotics.github.io/ironmind/>

Abstract

Egocentric human video offers a scalable data source for dexterous manipulation, yet using it to train humanoid robots presents two challenges: (1) an *embodiment gap*, as human hands differ structurally from robot end-effectors and low-cost egocentric recordings lack the torso kinematics required by conventional retargeting; and (2) *heterogeneous data quality*, including noisy hand-pose tracking and weakly aligned text annotations. We introduce **IronMind**, a vision-language-action (VLA) model that uses egocentric human video and heterogeneous robot data to pretrain policies for humanoid dexterous manipulation. To bridge the embodiment gap, IronMind bypasses explicit body-retargeting by using a *camera-space action representation*, the native reference space of egocentric video, and semantically aligning robot and human action dimensions. To mitigate data noise, we curate a pretraining corpus totaling more than 10,000 hours of egocentric human video and heterogeneous robot data using rule-based filtering, atomic task re-annotation, and per-frame quality weighting. Architecturally, IronMind uses a Mixture-of-Transformers policy that couples vision-language understanding with a flow-matching action expert. We also explore optional world-prior supervision from semantic features, spatial-geometric representations, and temporal world-model predictions. Across total pretraining budgets from 250 to 10,000 hours, validation loss decreases approximately log-linearly with data scale. Larger pretraining budgets also improve out-of-distribution real-robot manipulation after post-training: across six challenging tasks with unseen objects, affordances, and reasoning prompts, the 10,000-hour model achieves a 55.0% success rate, compared with at most 11.7% for every pretraining budget up to 5,000 hours and 5.0% without pretraining. At the same pretraining budget, the camera-space action representation also outperforms the torso-frame baseline. Together, these findings support pretraining with a camera-space action representation on large-scale human egocentric data as a scalable foundation for humanoid robot manipulation.

1 Introduction

Robot manipulation has advanced rapidly [Chi et al., 2023; Octo Model Team, 2024; Liu et al., 2025], driven by vision-language-action (VLA) models built on pretrained vision-language backbones that follow open-vocabulary instructions and generalize semantically across objects and scenes [Brohan et al., 2023; Kim et al., 2024; Black et al., 2025; NVIDIA, 2025b], achieving high success rates in structured tabletop and household tasks [Physical Intelligence, 2025a; Jiang et al., 2025]. However, scaling these models to general dexterous manipulation—defined as a unified policy that handles contact-rich, high-degree-of-freedom hand-object interactions—remains limited by data availability. Demonstrations of dexterous manipulation on real robots are slow and expensive to collect [Open X-Embodiment Collaboration, 2024; AgiBot-World-Contributors, 2025], and robotic manipulation lacks a web-scale data corpus.

Egocentric human video provides a low-cost data source for manipulation pretraining [Li et al., 2025a; Zheng et al., 2026; Li et al., 2026a], with 3D hand poses and camera motion estimated from head-mounted recordings. However, using egocentric video presents two primary challenges: (1) an *embodiment gap*: human hands differ structurally from robot end-effectors in kinematics and joint count, making direct retargeting inaccurate; and (2) a *quality gap*: hand-motion estimates contain tracking noise, text prompts are weakly aligned with the corresponding actions, and uncured video includes segments without manipulation, reducing data efficiency [Lin et al., 2025a; Belkhale et al., 2023; Hejna et al., 2024]. Addressing these two gaps is critical for effectively pretraining humanoid robot policies.

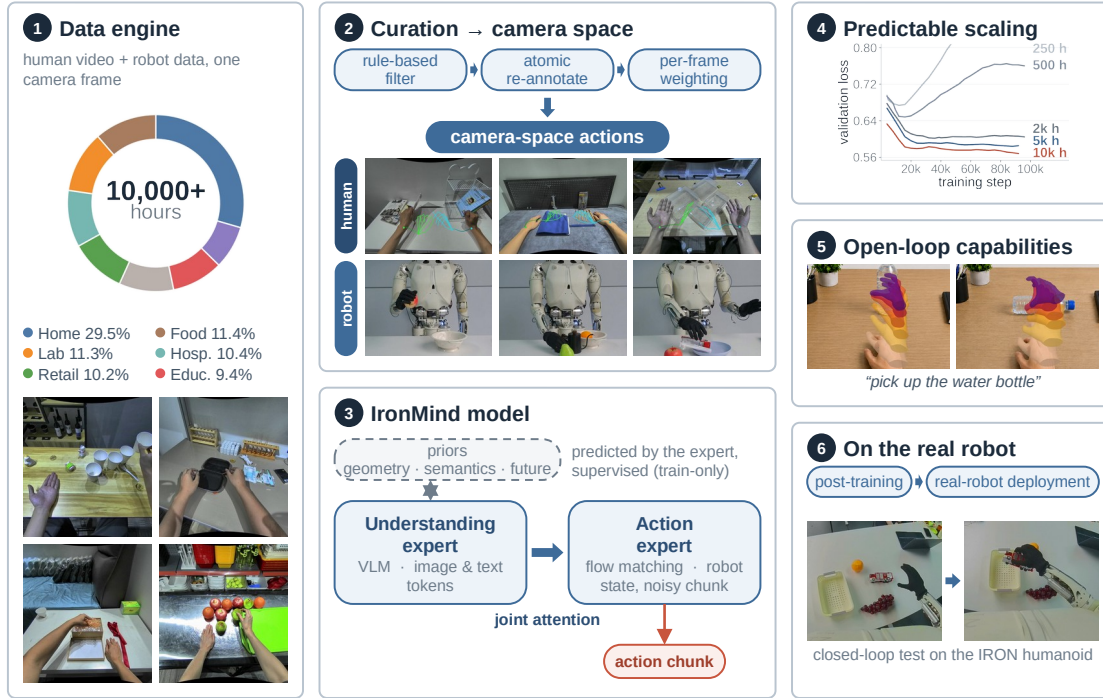


Figure 1: **Overview of IronMind.** (1) Pretraining Corpus: Over 10,000 hours in total, combining egocentric human video with heterogeneous robot data from non-target embodiments. (2) Curation & Representation: Data filtered, re-annotated, and unified into a shared camera-space action representation. (3) Architecture: A Mixture-of-Transformers policy with flow-matching action experts and optional multimodal priors. (4) Scaling: Validation loss scales log-linearly with pretraining hours without saturation up to 10,000 hours. (5) Zero-shot Evaluation: Open-loop trajectory prediction from a single unseen image and instruction. (6) Real-Robot Deployment: Closed-loop policy execution on IRON-R01.

In this paper, we introduce **IronMind**, a vision-language-action model pretrained with a camera-space action representation on large-scale egocentric human video for humanoid dexterous manipulation (Figure 1). To address the embodiment gap, IronMind avoids human-to-robot body retargeting. Egocentric video supports the estimation of camera-space trajectories (camera motion via SLAM and hand motion relative to camera space) rather than the wearer’s global body pose. Projecting these trajectories into a robot’s joint space requires assumptions about unobserved body kinematics (such as a rigid spine model), introducing label noise. IronMind instead expresses human and calibrated robot trajectories in the same camera-space action representation and aligns action dimensions semantically across embodiments [Yuan et al., 2026a]. To address the quality gap, a multi-stage curation pipeline applies rule-based filtering, atomic task re-annotation, and per-frame quality weighting to the pretraining video data.

Furthermore, dexterous manipulation requires predicting scene dynamics during hand–object contact. IronMind optionally incorporates world-model priors during pretraining by using learnable query tokens to condition a frozen video generation model [Team Wan, 2025] for future-frame prediction [Ma et al., 2026a]. This supervision is combined with semantic visual features [Siméoni et al., 2025] and spatial-geometric representations [Lin et al., 2025c]. The policy architecture uses a Mixture-of-Transformers [Liang et al., 2025] with separate vision-language and flow-matching action experts [Lipman et al., 2023; Black et al., 2025] linked via shared attention, preserving vision-language capabilities during action learning. All world-model prediction modules are removed after pretraining. This leaves only the learned query tokens and avoids auxiliary teacher-model computation at inference. Model performance is evaluated using camera-space open-loop metrics, policy-in-the-loop rollouts in held-out scenes, and real-robot testing.

We summarize our main contributions as follows:

1. **A camera-space pretraining recipe for humanoid manipulation.** We use a camera-space action representation as a unified human–robot interface, combining semantic alignment of action dimensions across embodiments with a multi-stage curation pipeline (rule-based filtering, atomic re-annotation, per-frame quality weighting) to process a total of over 10,000 hours of egocentric human video and heterogeneous robot data (Section 3.1). Validation loss decreases log-linearly with the pretraining data budget.
2. **A VLA architecture with optional multimodal world priors.** We introduce a dual-expert Mixture-of-Transformers backbone with flow-matching action decoding and explore optional world-prior supervision through future-frame prediction using a video generation model alongside semantic [Siméoni et al., 2025] and spatial-geometric representations [Lin et al., 2025c], without running the auxiliary teacher models at inference (Sections 3.2–3.3).
3. **An open-to-closed-loop evaluation suite.** We present an evaluation benchmark that combines camera-space open-loop approach and grasping metrics, policy-in-the-loop rollouts in unseen scenes, and real-robot evaluation protocols to assess prediction quality, sequential drift, and real-robot execution (Section 4).
4. **Empirical validation on robots.** On a real humanoid robot, larger pretraining budgets yield substantial improvements across six challenging out-of-distribution tasks with unseen objects, affordances, and reasoning prompts. The 10,000-hour model achieves a 55.0% success rate, representing a 43-percentage-point gain over smaller pretraining budgets (average success rates are at most 11.7% for all budgets up to 5,000 hours); at the same budget, the camera-space action representation also outperforms the torso-frame baseline (55.0% vs. 26.7%) (Section 4).

2 Related Work

Building robot policies on pretrained vision-language backbones [Bai et al., 2025; Qwen Team, 2026a] has become a standard approach to generalist manipulation. Early architectures cast robot control as the prediction of discrete text tokens [Brohan et al., 2023; Kim et al., 2024], with later work improving efficiency through learned action tokenization [Pertsch et al., 2025; Galaxea Team, 2026]. Other designs use continuous action heads based on diffusion or flow matching [Chi et al., 2023; Octo Model Team, 2024; Liu et al., 2025; Black et al., 2025]. These approaches underpin a range of large-scale VLA systems for real-world robot control, including the π series [Physical Intelligence, 2025a;b; 2026], GR00T N1 [NVIDIA, 2025b], Galaxea G0/G0.5 [Jiang et al., 2025; Galaxea Team, 2026], Qwen-VLA and Qwen-RobotManip [Wang et al., 2026a; Yuan et al., 2026a], LingBot-VLA [Wu et al., 2026a], InternVLA-A1/A1.5 [Cai et al., 2026; Ma et al., 2026a], Hy-Embodied-0.5 [Zhang et al., 2026], and Capek 0.5 [Chen et al., 2026d], with related formulations also explored in autonomous driving [NVIDIA, 2025a; Huang et al., 2026]. Parallel work investigates unified generation-action interfaces [Wang et al., 2025; Hu et al., 2026], adaptive reasoning [Lin et al., 2025b; Huang et al., 2025], memory mechanisms [Torne et al., 2026; Li et al., 2025b], and minimal policy formulations [Goyal et al., 2025]. IronMind builds on these advances, focusing on scalable data representations and a multi-stage pretraining recipe for learning humanoid manipulation from egocentric human video.

Learning manipulation from egocentric human data. Large-scale egocentric corpora [Grauman et al., 2022; 2024; Damen et al., 2022; Goyal et al., 2017] provide manipulation experience at a scale difficult to achieve with robot fleets alone, while recent datasets add accurate 3D hand-pose annotations [Hoque et al., 2025]. Building on these resources, systems such as VITRA, EgoScale, and EgoWAM demonstrate the effectiveness of egocentric pretraining at progressively larger scales [Li et al., 2025a; Zheng et al., 2026; Li et al., 2026a]. A central design choice is how to represent actions for transfer from humans to robots. Existing approaches include latent actions learned from video [Ye et al., 2025; Luo et al., 2026], explicit camera-space hand trajectories [Li et al., 2025a], relative wrist motion retargeted into robot joint space prior to training [Zheng et al., 2026], robot trajectories synthesized via cross-domain translation [Wang et al., 2026b; Chen et al., 2026b; Lin et al., 2026b], and joint human–robot representations [Zhu et al., 2026; Zha et al., 2026; Li et al., 2026b]. A complementary approach avoids explicit conversion between embodiments by retaining data from heterogeneous sources in their native action spaces and aligning their action dimensions semantically [Yuan et al., 2026a; Team, 2026]. Beyond action representation, growing evidence highlights the importance of data quality and diversity alongside raw data volume [Lin et al., 2025a; Belkhale et al., 2023; Hejna et al., 2024], while specialized tools facilitate data conversion [Liu et al., 2026]. IronMind follows the same principle of avoiding body retargeting: human demonstrations

use a camera-space action representation, while robot trajectories are transformed into the same camera space using directly measurable calibration parameters. Remaining differences in action dimensions are addressed through semantic alignment [Yuan et al., 2026a], enabling a single output head to learn from both human and robot action supervision without body retargeting or a learned latent bottleneck.

Integrating video prediction with control is an active area of policy learning. Existing approaches include jointly modeling physical dynamics and actions [Ma et al., 2026b; Li et al., 2026c; Guo et al., 2026; Wei et al., 2026], using predicted future frames as policy context [Ye et al., 2026; Yuan et al., 2026b; Li et al., 2026d], distilling latent world representations into the policy network [Sun et al., 2026; Chen et al., 2026c;a], scaling world-action pretraining on egocentric video corpora [Luo et al., 2026; Li et al., 2026a], and building omnimodal world models [NVIDIA, 2026]. Two findings from prior work guide our design. First, aligning representations of future states with action supervision helps make them useful for policy execution [Qiu et al., 2026; Chen et al., 2026c]. Second, pretrained video generation models encode rich priors about 3D dynamics and spatial structure [Wu et al., 2026b; Hu et al., 2025; Peebles and Xie, 2023; Team Wan, 2025]. The closest architectural precedent is InternVLA-A1.5, which encodes task-relevant future visual states in latent foresight tokens supervised by a frozen pretrained video generator. This design incorporates dynamics priors without requiring pixel-level video synthesis at deployment [Ma et al., 2026a]. IronMind adopts this approach, using world models solely to provide priors during training while leaving the video generator inactive at inference. We combine these predictive representations with semantic and geometric feature distillation [Siméoni et al., 2025; Lin et al., 2025c] within a Mixture-of-Transformers backbone [Liang et al., 2025], an architecture that combines expert-specific parameters with shared attention and has been applied to multimodal learning and embodied control [Huang et al., 2025; 2026].

Scaling and evaluation for robot learning. Scaling laws are well established for language models [Kaplan et al., 2020]; corresponding relationships in robot learning are being investigated through cross-embodiment datasets [Open X-Embodiment Collaboration, 2024; AgiBot-World-Contributors, 2025], empirical studies of dataset size, composition, and quality [Lin et al., 2025a; Hejna et al., 2024; Belkhale et al., 2023; Lin et al., 2026a], and analyses of action chunking strategies [Lazzati et al., 2026; Zhao et al., 2023]. Complementary post-training methods, such as reinforcement learning and physical interventions, further enhance real-world policy performance [Xu et al., 2026; Fei et al., 2026; Physical Intelligence, 2025b; Xiao et al., 2026]. Evaluation practices, however, remain fragmented: studies often rely on isolated offline metrics or success rates on a limited set of real-robot tasks, a limitation systematically analyzed by VIGIL [XPeng Robotics, 2026]. To address this fragmentation, we introduce an open-to-closed-loop evaluation suite that combines camera-space open-loop approach and grasp metrics, policy-in-the-loop rollouts in held-out scenes, and standardized real-robot testing protocols. Organized around an explicit taxonomy of capabilities, the suite supports systematic reporting of both successes and failure modes.

3 IronMind: The Framework

IronMind consists of three core components, as illustrated in Figure 1. The **pretraining data engine** (Section 3.1) processes diverse egocentric human videos and heterogeneous robot demonstrations to address differences in embodiment and data quality. It uses a shared camera-space action representation for trajectories from different embodiments and applies a multi-stage curation pipeline to filter data affected by tracking failures, dropped frames, and kinematically invalid hand motions, while refining inaccurate text annotations. The **model architecture** (Section 3.2) combines a vision-language-action (VLA) policy with optional multimodal priors, using a Mixture-of-Transformers [Liang et al., 2025; Ma et al., 2026a] that links a vision-language expert and a flow-matching action expert [Lipman et al., 2023; Black et al., 2025] via shared attention. During training, the backbone can receive auxiliary supervision from semantic features [Siméoni et al., 2025], spatial-geometric representations [Lin et al., 2025c], and future-frame predictions from a frozen video generation model [Team Wan, 2025]; the auxiliary prediction heads are removed at inference, leaving only a compact set of learned query tokens and avoiding the cost of running the auxiliary prediction branches during deployment. The **training recipe** (Section 3.3) consists of large-scale pretraining with the camera-space action representation, followed by post-training on a small curated dataset of real-robot teleoperation demonstrations, with robot action trajectories transformed into the same camera space using extrinsic calibration. Together, these components maintain a consistent action interface from heterogeneous pretraining data through adaptation and deployment on the target robot.

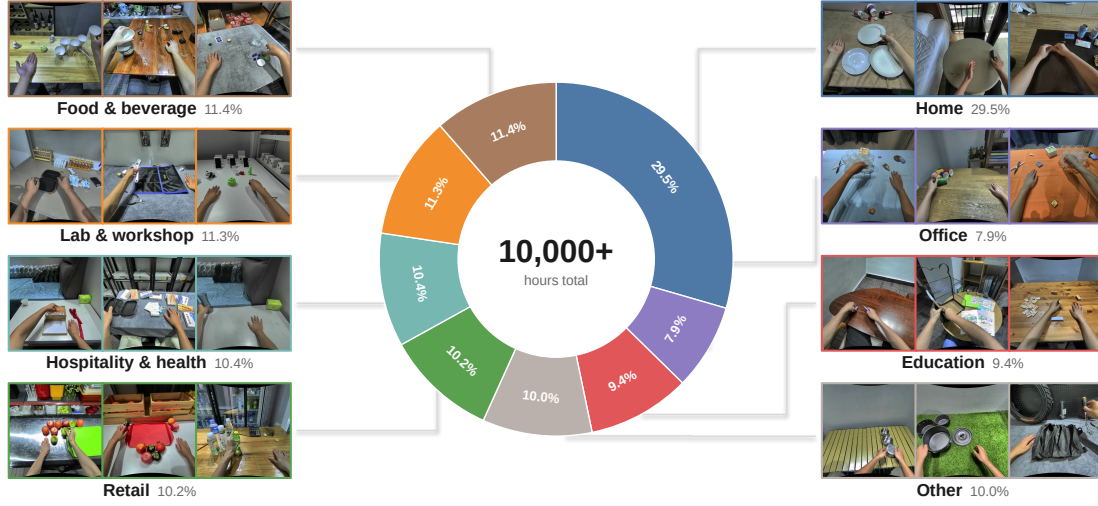


Figure 2: **Scene distribution of the IronMind pretraining corpus.** The dataset comprises over 10,000 total hours of egocentric human video and heterogeneous robot manipulation data spanning diverse real-world environments, all expressed using a camera-space action representation. The donut chart shows the proportion of recording hours in each scene category, alongside representative frames. Home environments account for the largest share; the remaining data cover food and beverage settings, laboratories and workshops, hospitality venues, retail spaces, educational facilities, offices, and other environments.

3.1 Pretraining data engine

3.1.1 Camera-space action representation

Recent work highlights the scalability of egocentric human video and the effectiveness of the camera-space action representation [Zheng et al., 2026; Li et al., 2026a; 2025a]. In standard robot control, the torso frame is a natural reference for upper-limb manipulation. Egocentric video, however, records motion from the camera’s perspective. Because low-cost recording hardware does not track the torso or spine, mapping human motion into a canonical robot torso frame requires assumptions about unobserved body kinematics (e.g., a rigid, vertical spine), which can introduce substantial retargeting noise [Li et al., 2025a; Yuan et al., 2026a].

We therefore bypass body retargeting by using a *camera-space action representation* [Yuan et al., 2026a; Team, 2026; Li et al., 2025a]. In addition to avoiding artifacts introduced by body retargeting, the camera-space action representation reduces sensitivity to *camera mounting misalignments* on physical robots. On physical humanoid robots, small deviations in head-camera mounting can create mismatches between visual observations and actions expressed in a torso frame. Expressing observations and actions in the same camera space helps reduce these mismatches.

For human data, parametric 3D hand poses and trajectories are extracted in the recording camera space [Ferguson et al., 2025]. For robot data, measured trajectories are transformed into the onboard head-camera space using extrinsic calibration. These robot demonstrations come from heterogeneous non-target embodiments; no IRON-R01 demonstrations are included during pretraining. To bridge structural differences between human hands and robot end-effectors, we use a unified action representation [Yuan et al., 2026a] that aligns action dimensions by their meaning across embodiments. Wrist and finger motions map to shared target-robot action dimensions. This alignment is semantic rather than kinematic: after the camera-space transformations described above, corresponding motion quantities occupy consistent output dimensions even when embodiments differ in joint structure. The policy can therefore learn a single action interface across data sources.

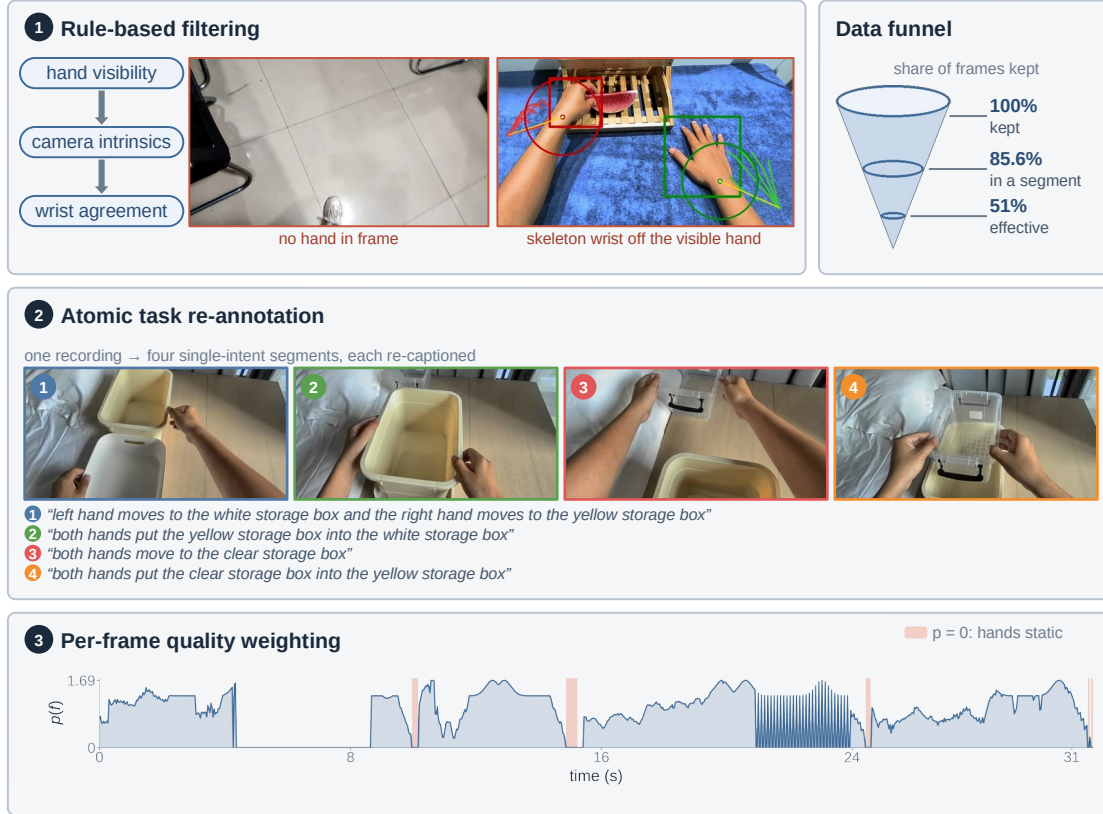


Figure 3: **Overview of the data curation pipeline with representative examples.** Rule-based filtering rejects recordings that fail the quality criteria (two examples shown). Atomic Re-annotation: Retained recordings are segmented into clips, each corresponding to a single task intent, and assigned refined captions. Per-frame Quality Weighting: Frames receive quality scores according to Eq. 1 (plotted for a sample recording) to determine their sampling weights during training.

3.1.2 Data curation pipeline

The shared action representation addresses embodiment mismatch, but it does not correct noisy trajectories or inaccurate language annotations. Effective scaling of egocentric pretraining depends on the availability of correctly annotated, kinematically valid action trajectories, not simply on total video duration [Zheng et al., 2026]. Raw egocentric recordings often contain long periods of passive observation, mismatches between captions and actions (e.g., clips labeled “pick up mug” that instead show an object being released), and coarse task descriptions that do not distinguish individual sub-actions. To address these issues, we process all video streams through a unified multi-stage curation pipeline (Figure 3).

Rule-based filtering. We first apply rule-based checks to remove segments affected by perception and tracking failures. We reject frames with missing or invalid camera intrinsics or implausible 3D hand poses caused by hand-pose estimation errors or severe occlusions.

Atomic task annotation. Coarse or missing captions provide insufficient supervision for policies that must follow fine-grained instructions. We segment continuous recordings into *atomic task* units, each consisting of a short manipulation segment with a single intent, and caption each segment using a vision-language model (Qwen3.6-35B-A3B [Qwen Team, 2026b]) to describe the objects and actions involved. We mask action losses at segment boundaries to prevent training windows from mixing supervision across task intents.

Kinematic velocity alignment. Human hand movements in video are typically faster and more agile than the slower, smoother motions in robot teleoperation data. To reduce this difference, we compute the mean

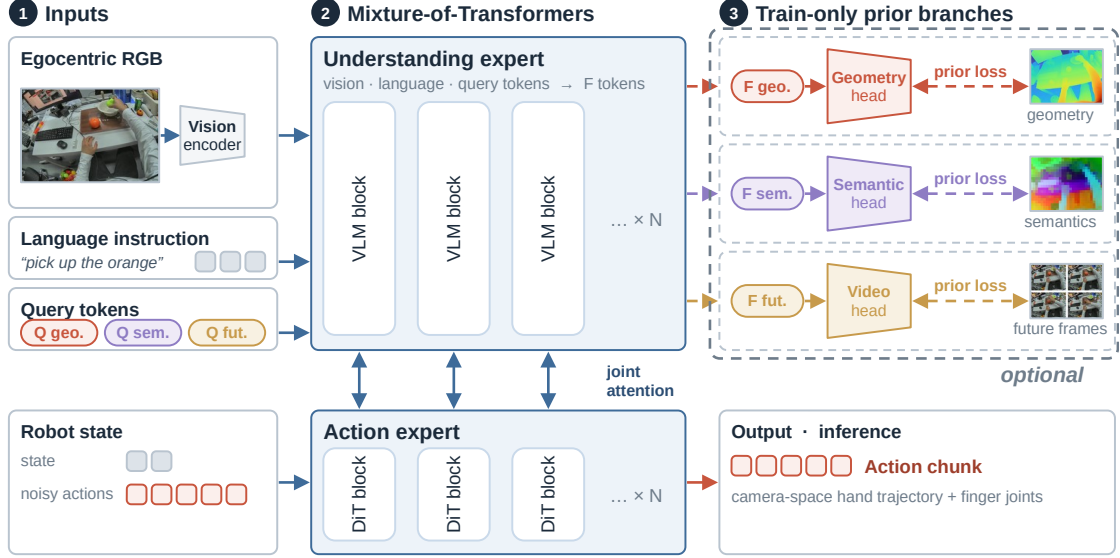


Figure 4: **IronMind architecture.** A vision-language understanding expert takes an egocentric image, a language instruction, and optionally a set of query tokens as input. During training, the query tokens produce world prior features (geometry, semantics, future frames) and are supervised using targets from pretrained models in each domain; the prior branches and prediction heads are removed at inference. The action expert attends to the understanding expert at shared attention layers and generates actions in the camera-space representation through flow-matching denoising, conditioned on robot state and action noise.

velocity for each source dataset split and apply temporal rescaling to better match human trajectory speeds to those of real-robot teleoperation data.

Per-frame quality weighting. In addition to binary filtering, we assign sampling weights to candidate frames within atomic segments using continuous quality measures. For each candidate frame f , a quality score $p(f)$ is computed over a short trailing window:

$$p(f) = e(f)g(f) \cdot s_{\text{cam}}(f)s_{\text{move}}(f)b_{\text{vel}}(f) \in [0, C_{\text{max}}], \quad (1)$$

where $e(f)$ and $g(f)$ are binary gates for hand visibility and camera tracking stability, while $s_{\text{cam}}(f)$, $s_{\text{move}}(f)$, and $b_{\text{vel}}(f)$ encode camera rotational stability, active hand translation, and a bonus for pauses at segment boundaries, respectively.

The dataloader draws training samples using the cumulative distribution induced by $p(f)$ via inverse-transform sampling. This prioritizes frames with smooth motion and clear manipulation intent while retaining neighboring frames for context and avoiding costly dataset regeneration.

The stages serve distinct roles: filtering determines which observations are valid, atomic annotation defines the intended behavior, temporal rescaling reduces source-dependent speed differences, and quality weighting controls how often valid frames are sampled.

3.2 Model Architecture

IronMind is a vision-language-action (VLA) policy that maps egocentric visual observations and open-vocabulary language instructions to smooth, metric trajectories expressed using the camera-space action representation (Figure 4). To achieve scalable pretraining while preserving reasoning and instruction-following capabilities, the architecture is designed to balance low-level trajectory regression with high-level vision-language understanding.

IronMind addresses this via two core principles: **(1) Dual-Expert Decoupling via Mixture-of-Transformers (MoT):** using separate experts for perception and reasoning and for flow-matching action decoding, with selective shared attention connecting the two; and **(2) Multimodal Prior Distillation via Latent Queries:** using auxiliary dynamics, geometry, and semantics objectives only during training.

3.2.1 Dual-Expert Mixture-of-Transformers Architecture

Standard single-stream VLA models append discrete or continuous action tokens directly to the vision-language context sequence. However, directly updating the vision-language trunk with gradients from low-level trajectory regression can interfere with its pretrained open-vocabulary representations. Following the Mixture-of-Transformers (MoT) paradigm [Liang et al., 2025; Ma et al., 2026a], IronMind decouples perception and reasoning from action generation into two dedicated transformer streams with separate parameters:

Understanding Expert. Initialized from a pretrained vision-language backbone, the understanding expert processes the head-camera RGB image and the natural-language instruction. It retains its native causal and cross-attention language decoding heads. This design helps preserve open-vocabulary grounding and instruction understanding during action pretraining.

Action Expert. Implemented as a Diffusion Transformer (DiT) variant trained with a flow-matching objective [Lipman et al., 2023; Black et al., 2025], the action expert models distributions over continuous action trajectories. During training, it conditions on state tokens and noisy action chunks to predict vector fields that transform noise into ground-truth action sequences. At inference, numerical integration of the predicted vector field over a small number of steps produces a temporally coherent action chunk. Predicting multi-step trajectories simultaneously encourages temporal smoothness and helps mitigate compounding execution errors during closed-loop rollout.

Shared Attention Fusion. IronMind connects the two experts through shared attention layers at selected depths [Ma et al., 2026a]. At designated fusion layers throughout the transformer stack, understanding and action tokens participate in joint attention over the concatenated sequence, with stream-specific projection matrices applied to each stream. This selective cross-stream interaction allows the action expert to access fine-grained visual features and language context directly from the understanding backbone, while reducing direct interference with the pretrained vision-language representations.

3.2.2 Multimodal World Priors Supervision

Robotic manipulation requires an understanding of depth, surface geometry, and changes in the scene during contact, all of which receive only indirect supervision from standard 3D trajectory regression. Following the latent-query formulation of InternVLA-A1.5 and EgoWAM [Ma et al., 2026a; Li et al., 2026a], IronMind optionally incorporates world priors during pretraining by adding a compact set of learnable *world query* tokens to the understanding expert’s prefix.

During training, the query tokens interact with the image and language context through self-attention in the understanding expert, producing compact latent representations of geometry, semantics, and future dynamics. These latent representations condition auxiliary heads that predict scene dynamics, geometry, and semantic features:

(1) **Spatial-Geometric Prior:** the queries condition a prediction head supervised by dense depth representations [Lin et al., 2025c], encouraging the model to associate image locations with metric depth; (2) **Semantic Representation Prior:** the queries condition a feature decoder that predicts semantic features from a self-supervised model [Siméoni et al., 2025], encouraging object representations that are less sensitive to appearance changes; and (3) **Dynamic World-Model Prior:** the latent queries condition a frozen video generation model [Team Wan, 2025] to predict future variational autoencoder (VAE) latents, encouraging the understanding expert to encode contact dynamics and spatiotemporal changes. We also evaluate 3D motion flow as an alternative supervision target for this branch.

All external pretrained feature extractors, depth estimators, and video generation models remain frozen throughout policy pretraining. The auxiliary-loss gradients enter the understanding expert through the learnable query-token pathways. After pretraining, we remove the auxiliary decoder heads and target-generation networks and no longer compute the auxiliary losses. At inference, the action expert accesses the learned representations through shared attention. The learned query tokens are retained, while the auxiliary prediction branches are not executed during deployment.

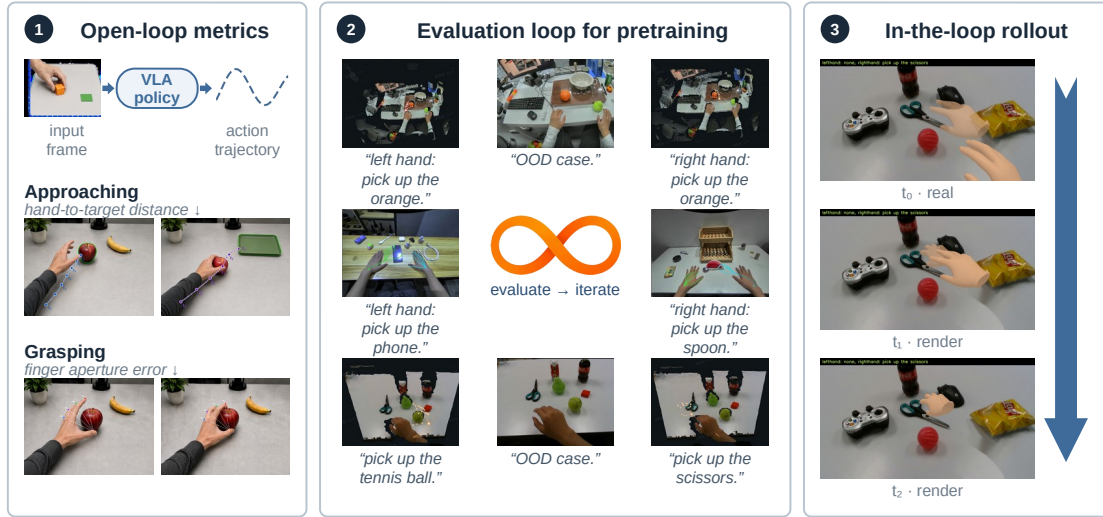


Figure 5: **Evaluation protocol.** **Left: Quantitative open-loop evaluation.** With the camera-space action representation, target approach is measured via d_{p10} and grasp execution via finger aperture error e_{FA} . **Middle: Evaluation loop for pretraining.** Held-out RealSense tabletop scenes with language prompts are used both to iterate pretraining and to score policy-in-the-loop rollouts. **Right: Policy-in-the-loop rollout.** Chunk t_0 is predicted from the observation; subsequent chunks t_1, t_2 re-render the predicted hand while the camera and physical scene stay fixed.

3.3 Training Paradigm

IronMind uses a two-stage training paradigm to transfer representations learned from heterogeneous pretraining data to the target robot. In the **large-scale pretraining stage**, we train the MoT policy primarily on large-scale egocentric human video, supplemented by heterogeneous robot teleoperation sequences from non-target embodiments. During this stage, we optimize the camera-space flow-matching action loss, optionally together with auxiliary objectives for depth, DINOv3 semantic features, and video dynamics. These world priors provide additional supervision during pretraining, encouraging the understanding expert to learn representations of scene geometry, semantics, and dynamics. Across the scaling experiments, the architecture, optimization recipe, and heterogeneous robot-data mixture remain fixed; additional experience is introduced through the human-video portion of the combined corpus. This design attributes differences across checkpoints primarily to data scale.

In the subsequent **post-training stage**, we adapt the pretrained policy to the target robot using a curated dataset of teleoperated trajectories. IRON-R01 demonstrations are introduced only at this stage. The action interface retains the camera-space action representation: we transform robot trajectories into head-camera space using extrinsic calibration, allowing us to reuse the pretrained understanding expert and action head without changing the action representation (a torso-frame baseline is trained only for the comparison in Section 4.5). We fine-tune both experts on real-robot data to adapt the pretrained policy to IRON-R01. Thus, post-training changes the embodiment while preserving pretrained action semantics.

4 Experiments

Evaluating a Vision-Language-Action (VLA) manipulation policy directly on a real humanoid robot is time-intensive, safety-critical, and throughput-constrained. Real-robot supervision is naturally sparse, and the control loop between model decisions and physical outcomes is long. To systematically navigate this trade-off, we evaluate IronMind through a multi-tiered empirical framework that trades interaction fidelity against throughput. Our investigation centers on four primary dimensions: verifying policy capability scaling over total pretraining budgets up to 10,000 hours, examining optional auxiliary multimodal world-model priors, testing the camera-space action representation against a torso-frame baseline, and establishing real-robot success on IRON-R01 under out-of-distribution (OOD) scenarios.

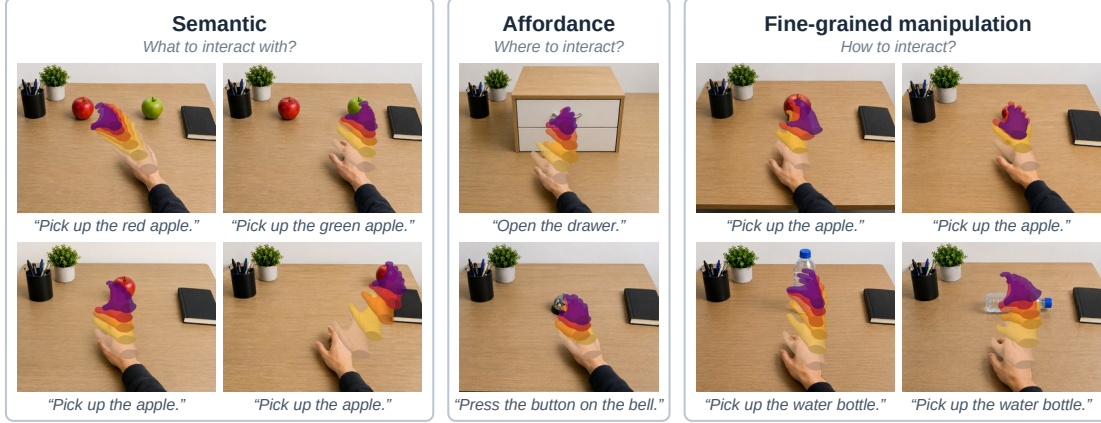


Figure 6: **Open-loop capability evaluation on OOD scenes.** Given an unseen RGB image and a text instruction, IronMind predicts open-loop hand trajectories (rendered hand meshes colored chronologically from light skin tone to purple). Visualizations illustrate fine-grained spatial grounding, including adjustments of grasp aperture and wrist orientation to target geometry.

4.1 Implementation Details and Training Recipe

All IronMind models are trained with AdamW [Loshchilov and Hutter, 2019] using a global batch size of 1024 and a cosine learning-rate schedule that decays to 10% of the peak rate. The reported pretraining budgets combine egocentric human video with heterogeneous robot data and contain no IRON-R01 demonstrations. Post-training uses approximately 40,000 filtered teleoperated episodes collected on IRON-R01.

4.2 Evaluation Protocols

We adopt a hierarchical evaluation protocol with increasing interaction fidelity, spanning camera-space open-loop metrics, offline policy-in-the-loop rollouts, and real-robot execution (Figure 5).

The three tiers isolate complementary sources of error. Open-loop evaluation measures the quality of a single predicted chunk without feedback, whereas policy-in-the-loop rollouts expose error accumulation when predictions are repeatedly conditioned on the model’s own outputs. Real-robot evaluation then introduces factors absent from image-based rollouts, including controller error, camera motion, and physical contact. Using the same approaching metrics in the first two tiers makes their results directly comparable before moving to task-level success on IRON-R01.

4.2.1 Open-Loop Evaluation

Open-loop evaluation scores one predicted action chunk against the observed scene without executing it. We collect all open-loop evaluation sets, and every case starts from a real human hand reconstructed by the same front end that labels the pretraining data, so the observation and the proprioceptive state stay in one domain. We measure approaching and grasping quantitatively, probe individual capabilities with case studies, and run a user study on general actions.

Quantitative Metrics. We score approaching and grasping on two self-collected sets. **Approaching.** We instruct the model to reach for objects on a table-top. We capture 101 RGB-D snapshots of a table-top with a RealSense D435, rearranging its 6–15 everyday objects between shots, and annotate 23 object categories with captions and 3D point clouds segmented from the depth image. Each snapshot also contains one or two real human hands resting 25–50 cm from the objects. Pairing every hand with every unambiguous object in its snapshot gives 970 test cases (492 right-hand, 478 left-hand) under 46 prompts. Let $\mathcal{P} \subset \mathbb{R}^3$ be the target’s point cloud and $\mathbf{x}_{t,k}$ be the predicted positions of wrist and fingers. The per-step hand-to-target distance is $d_t = \min_k \min_{\mathbf{p} \in \mathcal{P}} \|\mathbf{x}_{t,k} - \mathbf{p}\|_2$, and we report three summaries of it over the chunk: d_{p10} , the 10th percentile of $\{d_t\}$ in cm, a robust closest-approach measure that is less

Table 1: **Open-loop performance vs. pretraining budget** (one chunk).

Budget	d_{p10} (cm) ↓	Prog. (%) ↑	θ_{align} (°) ↓	e_{FA} (mm) ↓
250 h	37.8	14.7	47.8	19.97
500 h	33.4	28.1	41.7	19.54
2,000 h	32.0	30.7	36.0	23.83
5,000 h	30.8	34.0	41.2	20.33
10,000 h	25.8	44.9	26.0	15.66

Table 2: **Policy-in-the-loop performance vs. pretraining budget** (three chunks, 970 cases).

Budget	d_{p10} (cm) ↓	Prog. (%) ↑	θ_{align} (°) ↓
250 h	32.0	21.4	46.1
500 h	25.6	32.2	39.5
2,000 h	23.0	39.0	33.9
5,000 h	22.7	39.1	36.3
10,000 h	14.6	49.3	21.6

sensitive to isolated steps; *progress*, $(d_0 - d_T)/d_0$, the share of the initial gap closed by the last step; and $\theta_{\text{align}} = \angle(\bar{\mathbf{p}} - \mathbf{x}_{0,\text{wrist}}, \mathbf{x}_{T,\text{wrist}} - \mathbf{x}_{0,\text{wrist}})$, the angle between the direction to the target centroid $\bar{\mathbf{p}}$ and the wrist’s net displacement, 0° meaning the hand heads straight for the target. Lower is better for d_{p10} and θ_{align} , higher for progress. **Grasping.** The grasping set contains 120 grasp events from 10 self-collected scenes, one target object each, with human-reviewed closure-onset anchors. Mean finger aperture is $a_t = \frac{1}{4} \sum_{j=1}^4 \|\mathbf{x}_{t,\text{thumb}} - \mathbf{x}_{t,j}\|_2$ over the four fingertips. The ground-truth segment from the anchor, where the hand starts to close, to full closure is resampled to the length of one action chunk, the policy predicts one chunk from the anchor observation, and e_{FA} is the mean absolute difference between the predicted and ground-truth aperture profiles over that chunk (mm), measuring how closely the predicted closure follows the demonstration.

Case Studies across Capabilities. Beyond the aggregate metrics, we probe individual capabilities with case studies on self-collected out-of-distribution scenes: semantic grounding (*what* to interact with), affordance (*where* to interact) and fine-grained manipulation (*how* to interact). For each case we render the predicted hand trajectory in 2D and 3D and inspect where each capability succeeds or fails (Figure 6).

User Study on General Actions. For general hand actions beyond reaching and grasping, quantitative metrics do not adequately capture whether a predicted movement is plausible, so we run a user study. For each held-out scene we prompt the models under comparison with the annotated instruction and render the predicted hand actions onto the frames. A trial shows the ground-truth demonstration next to the unlabeled, shuffled renderings, and the participant picks the better prediction or a tie. The study compares the 10,000 h and 250 h checkpoints on 20 held-out trials, each judged by 5 participants, yielding 100 votes. We compute win rate over decided votes and report ties in Table 3.

4.2.2 Policy-in-the-Loop Rollout

To probe sequential error accumulation over multi-chunk execution without running the policy on a robot, we run policy-in-the-loop rollouts on the same 970 approaching cases. We chain three relative chunks. After chunk k , the predicted hand is rendered back into the observation for chunk $k + 1$; camera pose and scene geometry stay fixed (Figure 5, right). We report the approaching metrics of the open-loop protocol, d_{p10} , progress and θ_{align} , over all 970 cases (Table 2). Because three chunks are long enough to close a substantial fraction of the 25–50 cm start gap, d_{p10} here is a reach score rather than a restatement of d_0 .

4.2.3 Closed-Loop Real-Robot Evaluation

Real-robot evaluations use an IRON-R01 unit distinct from the one used for post-training data collection, with the policy running in closed loop at 10 Hz under contact dynamics and sensor feedback.

4.3 Pretraining Experience Scaling

Across pretraining budgets from 250 to 10,000 hours, IronMind’s validation loss follows an approximately log-linear scaling law (Figure 7). The scaling fits a log-linear slope of -0.058 per decade of pretraining hours without exhibiting loss saturation up to 10,000 hours.

Larger pretraining budgets also improve open-loop spatial and grasping behavior (Table 1). Scaling pretraining data from 250 h to 10,000 h reduces d_{p10} from 37.8 to 25.8 cm (31.7%), increases approach progress from 14.7% to 44.9%, and lowers the finger-aperture error e_{FA} from 19.97 to 15.66 mm. The

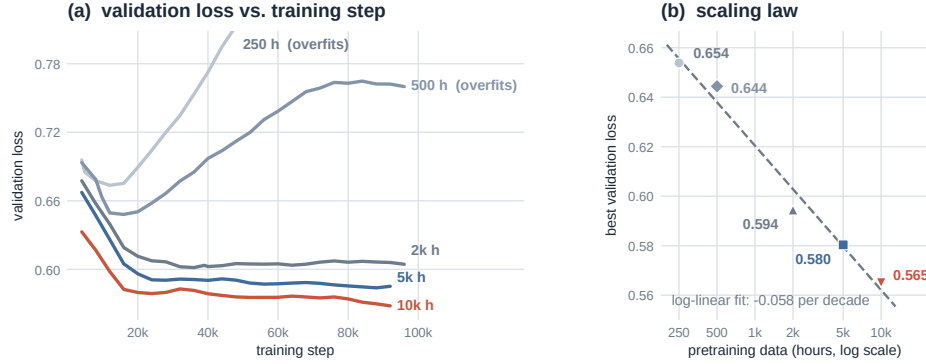


Figure 7: **Pretraining data scaling.** Pretraining validation loss exhibits a smooth log-linear scaling trend across total pretraining budgets (250 h to 10,000 h) with a slope of -0.058 per decade of hours, with larger budgets also improving open-loop target approach and grasp closure.

intermediate budgets are not strictly ordered by every metric, but target approach improves monotonically with data, and the largest budget is best in all four columns. The 10,000-hour checkpoint also performs best in policy-in-the-loop rollouts (Table 2): from 250 h to 10,000 h, progress rises from 21.4% to 49.3%, alignment falls from 46.1° to 21.6° , and d_{p10} from 32.0 to 14.6 cm, while 5,000 h and 2,000 h stay nearly tied (39.1% vs. 39.0% progress). In the blind user study on held-out egocentric frames (Table 3), raters prefer the 10,000 h pretrained checkpoint over the 250 h baseline in 47 of the 70 decided votes, a 67.1% **win rate**; the remaining 30 of the 100 votes are ties.

Table 3: **Human preference win rate vs. pretraining data budget.** The blind user study in Section 4.2.1 covers 20 held-out trials: 100 votes: 30 ties and 70 decisions; win rate is computed over decided votes.

Pretraining Budget	Votes	Win Rate $\uparrow$
250 h	23	32.9%
10,000 h	47	67.1%

4.4 World Priors Ablation

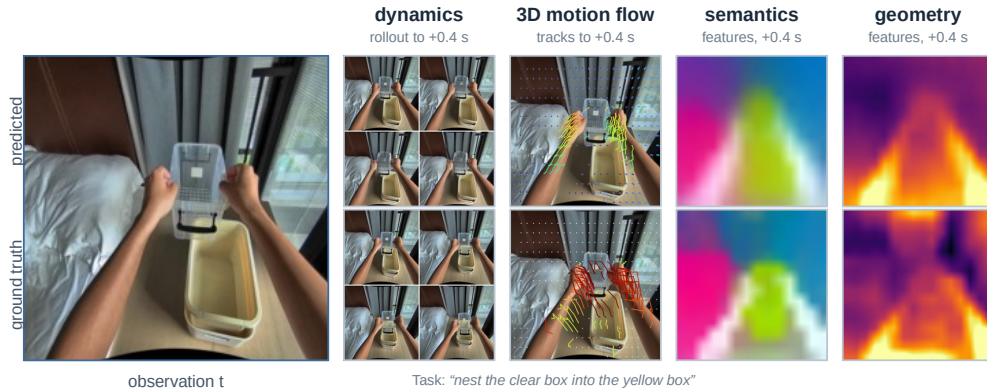


Figure 8: **What the world prior branches predict from one observation.** The left panel shows observation t ; for each branch the prediction (top) sits above the ground truth (bottom) in the same rendering. The panels show future-frame dynamics, 3D motion flow, semantic features, and geometry targets over a 0.4 s horizon.

Table 4: **World-prior ablation on the open-loop approaching set.** Results are averaged over several checkpoints from each pretraining run. Lower is better for d_{p10} and θ_{align} ; higher is better for progress.

Prior Configuration	d_{p10} (cm) ↓	Prog. (%) ↑	θ_{align} (°) ↓
<i>Reference configurations</i>			
no world-prior supervision (baseline)	27.6	40.9	34.8
three priors (default: future target, 0.4 s)	22.6	53.4	26.5
<i>Prior composition</i>			
without the semantics prior	24.1	50.2	29.2
without the depth prior	25.7	46.1	31.1
without the dynamics prior	24.1	51.1	26.8
<i>Dynamics supervision</i>			
target: current frame	27.3	42.7	33.3
horizon 0.2 s	25.8	45.7	30.5
horizon 0.8 s	24.8	49.3	28.3
dynamics target: 3D motion flow	25.2	47.1	28.6

All arms use the same pretraining recipe and corpus and differ from the three-prior configuration by a single configuration change. The three priors are the semantic, depth, and dynamics branches described in Section 3.2.2, supervised by DINOv3, Depth-Anything-3-Base, and the frozen Wan2.2 video generator, respectively. Table 4 reports results on the 970-case approaching set described in Section 4.2.1. Each entry is averaged over several checkpoints from the same training run. Because the differences are only a few centimeters, we interpret them as tendencies rather than conclusive effects. One arm replaces the frozen video generator with an EgoWAM-style 3D motion-flow head [Li et al., 2026a]. As illustrated in Figure 8, this flow-matching decoder predicts the camera-space displacement of query points over the prediction horizon, using offline depth and tracking labels for supervision.

The full three-prior configuration performs best across all three reported metrics. Relative to the no-prior baseline, the full configuration reduces d_{p10} from 27.6 to 22.6 cm, increases progress from 40.9% to 53.4% and improves alignment from 34.8° to 26.5° (Table 4). Removing any one branch increases d_{p10} by 1.5–3.1 cm, with the largest change observed when depth is removed, which is consistent with contributions from all three branches. Replacing future prediction with current-frame reconstruction largely reverses the gain (27.3 cm, 42.7%), consistent with future targets providing useful information beyond current-frame feature supervision. The default horizon of approximately 0.4 s also performs better than 0.2 or 0.8 s, possibly reflecting a trade-off between predictive information and uncertainty at longer horizons. At this training budget, the frozen video generator shows a modest advantage over the 3D motion-flow head as a dynamics target; the sparser flow labels may partly explain this difference.

4.5 Real-Robot Experiments

We evaluate IronMind in closed loop on IRON-R01 at 10 Hz. To test zero-shot generalization and spatial robustness, we use out-of-distribution (OOD) settings with object instances, affordances, and reasoning prompts not seen during post-training (Figure 9). For each setting, we report binary task success and a milestone-based progress score (25%, 50%, 75%, or 100% completion) over 10 trials. Success requires completion of the full instructed task, whereas progress credits completed stages such as reaching, grasping, transporting, or releasing. Reporting both distinguishes immediate failures from executions that complete most of the sequence but miss the final condition.

We categorize real-robot out-of-distribution (OOD) evaluations into three groups: **semantic generalization**, **affordance adaptation**, and **spatial reasoning**. Semantic generalization tests grounding of unseen object instances and appearances, such as placing an orange or a toy car into a basket under visual clutter. Affordance adaptation tests interactions that require unseen grasp strategies, such as grasping a kettle by its handle rather than its body, or lifting a basket by its handle to hand it to a person. Spatial reasoning tests relational and attribute-based instructions (e.g., “put the grapes in the left basket” or “pick up the blue cup rather than the yellow cup”).

Real-robot results on IRON-R01 are consistent with the pretraining scaling trends observed in open-loop evaluation. Across OOD task categories (Figure 10), the average success rate stays below 12% for every

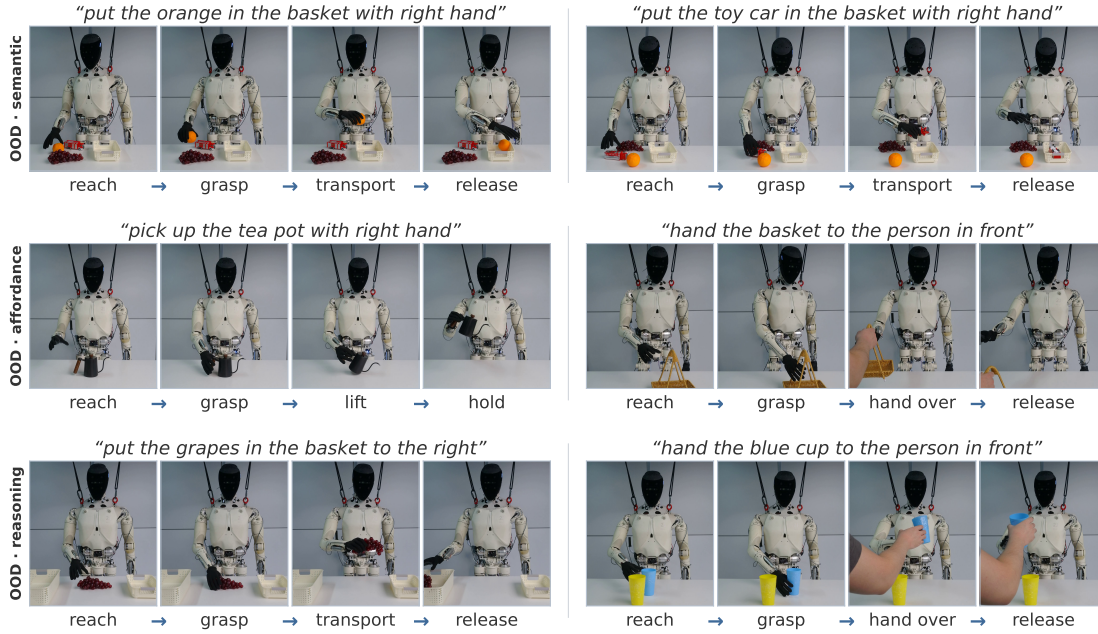


Figure 9: **Real-robot out-of-distribution (OOD) evaluation on IRON-R01.** Success and progress rates are evaluated across three out-of-distribution categories (Semantic, Affordance, and Reasoning OOD).

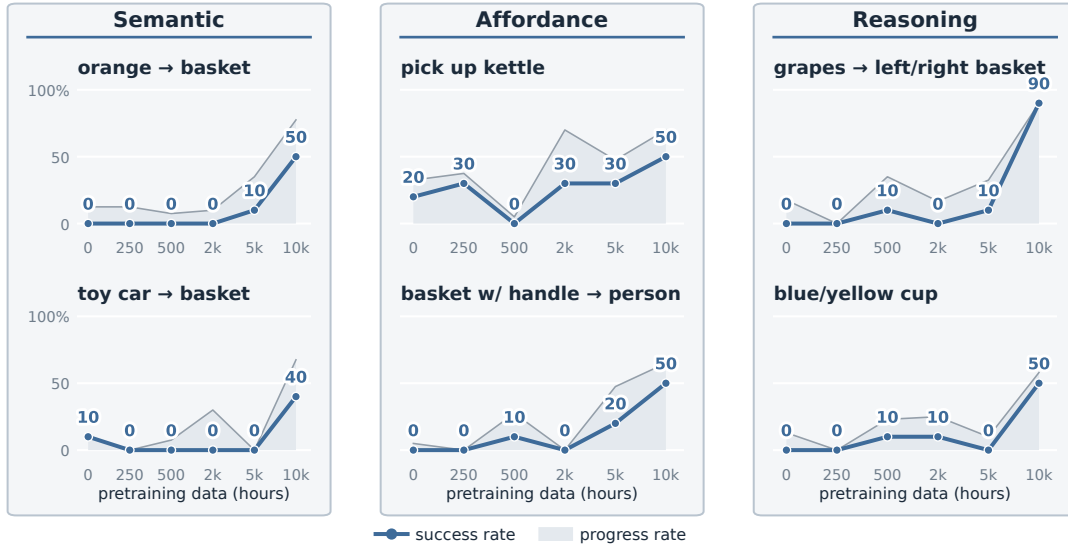


Figure 10: **Scaling effect on real-robot OOD success and progress rates.** Success (blue) and milestone progress (gray) are reported for six OOD tasks grouped into Semantic, Affordance, and Reasoning categories. At smaller budgets (0–2,000 h), performance is generally low and non-monotonic across tasks. The clearest improvement occurs between 5,000 and 10,000 h, when success increases on all six tasks—for example, from 10% to 90% for grapes to the left/right basket and from 0% to 50% for the blue/yellow cup—showing an overall positive scaling trend despite fluctuations at smaller scales.

training budget up to 5,000 hours and rises above 40% (40–90% per task) at 10,000 hours. The increase is strongest on the Reasoning and Semantic OOD tasks and weaker on the Affordance OOD tasks. To isolate the effect of data scale, these experiments do not use world-prior supervision.

Table 5: **Real-robot ablations.** Results are averaged over the six OOD tasks of Figure 9 (10 trials each), using the same post-training data and compute budget. Parentheses report percentage-point drops relative to the first row.

Model variant	Success $\uparrow$	Progress $\uparrow$
IronMind (ours, camera-space, no world-prior supervision)	55.0%	71.3%
IronMind (ours, torso-frame baseline)	26.7% ($\downarrow$ 28.3)	46.2% ($\downarrow$ 25.1)
IronMind (ours, with world-prior supervision)	31.7% ($\downarrow$ 23.3)	53.3% ($\downarrow$ 18.0)
InternVLA-A1.5 [Ma et al., 2026a]	30.0% ($\downarrow$ 25.0)	47.8% ($\downarrow$ 23.5)

We also compare IronMind variants with the external VLA baseline InternVLA-A1.5 [Ma et al., 2026a]. We take the public weights from InternVLA-A1.5 and post-train it on the same post-training data as IronMind. We then compare it with the 10,000-hour IronMind model on the same OOD tasks as in Figure 9. The results are shown in Table 5. IronMind (with the camera-space action representation and without world-prior supervision) obtains the highest overall success and progress rates under this evaluation setup. However, IronMind uses a shared unified action space for pretraining and post-training, while the action space is re-initialized for InternVLA-A1.5.

Ablation experiments comparing different variants of IronMind (Table 5) show two patterns regarding representation space and supervision objectives. First, the camera-space action representation performs better than the torso-frame baseline. To ensure a rigorous and fair comparison, both models were trained on the same 10,000-hour dataset (albeit transformed to their respective action representations). Each model used its assigned action representation consistently during both pretraining and post-training. These results suggest that a camera-space action representation can improve generalization in egocentric pretraining for humanoid robot manipulation. Although world-prior supervision improves some open-loop metrics, its real-robot variant only modestly outperforms InternVLA-A1.5 and remains below IronMind without world priors. This may reflect a mismatch between auxiliary prediction objectives and contact-rich closed-loop control.

5 Limitations

Several limitations remain. First, the corpus remains concentrated in homes, kitchens, and workshops, limiting extreme scene diversity. Second, the camera-space action representation reduces but does not eliminate target-robot dependence, still requiring camera extrinsic calibration and post-training on real-robot data. The scaling study covers one architecture and data mixture, while real-robot evaluation covers one embodiment, six OOD tasks, and 10 trials per task; longer-horizon and broader contact-rich tasks remain untested. Future work will further examine how gains from world-prior supervision transfer from open-loop evaluation to closed-loop deployment. Finally, comparisons beyond InternVLA-A1.5, including DreamZero [Ye et al., 2026], remain future work.

6 Conclusion

We presented IronMind, a VLA system pretrained on large-scale egocentric human and heterogeneous robot data for humanoid dexterous manipulation. Its unified camera-space action representation bridges human observation and robot execution without explicit embodiment retargeting. A curated data engine supports a dual-expert Mixture-of-Transformers policy with flow-matching action generation and optional multimodal world priors. Validation loss scales approximately log-linearly from 250 to 10,000 pretraining hours. Evaluation spans open-loop prediction, policy-in-the-loop rollouts, and closed-loop robot execution. After post-training, the 10,000-hour model achieves 55.0% success across six out-of-distribution tasks and outperforms the torso-frame baseline (55.0% vs. 26.7%). Together, these results support camera-space pretraining as a scalable foundation for humanoid manipulation. Future work will scale data and embodiments and explore longer-horizon tasks, reinforcement learning, and human intervention.

References

- AgiBot-World-Contributors. Agibot world colosseo: A large-scale manipulation platform for scalable and intelligent embodied systems, 2025. URL <https://arxiv.org/abs/2503.06669>.
- Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yanzhi Zhu, Mingkun Yang, Zhaoai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-vl technical report, 2025. URL <https://arxiv.org/abs/2502.13923>.
- Suneel Belkhale, Yuchen Cui, and Dorsa Sadigh. Data quality in imitation learning, 2023. URL <https://arxiv.org/abs/2306.02437>.
- Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Lucy Xiaoyang Shi, James Tanner, Quan Vuong, Anna Walling, Haohuan Wang, and Ury Zhilinsky. π_0 : A vision-language-action flow model for general robot control, 2025. URL <https://arxiv.org/abs/2410.24164>.
- Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, Pete Florence, Chuyuan Fu, Montse Gonzalez Arenas, Keerthana Gopalakrishnan, Kehang Han, Karol Hausman, Alexander Herzog, Jasmine Hsu, Brian Ichter, Alex Irpan, Nikhil Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, Isabel Leal, Lisa Lee, Tsang-Wei Edward Lee, Sergey Levine, Yao Lu, Henryk Michalewski, Igor Mordatch, Karl Pertsch, Kanishka Rao, Krista Reymann, Michael Ryoo, Grecia Salazar, Pannag Sanketi, Pierre Sermanet, Jaspiar Singh, Anikait Singh, Radu Soricut, Huong Tran, Vincent Vanhoucke, Quan Vuong, Ayzaan Wahid, Stefan Welker, Paul Wohlhart, Jialin Wu, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Tianhe Yu, and Brianna Zitkovich. Rt-2: Vision-language-action models transfer web knowledge to robotic control, 2023. URL <https://arxiv.org/abs/2307.15818>.
- Junhao Cai, Zetao Cai, Jiafei Cao, Yilun Chen, Zeyu He, Lei Jiang, Hang Li, Hengjie Li, Yang Li, Yufei Liu, Yanan Lu, Qi Lv, Haoxiang Ma, Jiangmiao Pang, Yu Qiao, Zherui Qiu, Yanqing Shen, Xu Shi, Yang Tian, Bolun Wang, Hanqing Wang, Jiaheng Wang, Tai Wang, Xueyuan Wei, Chao Wu, Yiman Xie, Boyang Xing, Yuqiang Yang, Yuyin Yang, Qiaojun Yu, Feng Yuan, Jia Zeng, Jingjing Zhang, Shenghan Zhang, Shi Zhang, Zhuoma Zhaxi, Bowen Zhou, Yuanzhen Zhou, Yunsong Zhou, Hongrui Zhu, Yangkun Zhu, and Yuchen Zhu. Internv1a-1: Unifying understanding, generation and action for robotic manipulation, 2026. URL <https://arxiv.org/abs/2601.02456>.
- Boyu Chen, Yi Chen, Lu Qiu, Jerry Bai, Yuying Ge, and Yixiao Ge. Unit: Toward a unified physical language for human-to-humanoid policy learning and world modeling, 2026a. URL <https://arxiv.org/abs/2604.19734>.
- Sijin Chen, Kaixuan Jiang, Haixin Shi, Yanhui Wang, Weiheng Zhong, Haosheng Li, Bo Jiang, Yuxiao Liu, and Xihui Liu. Translation as a bridging action: Transferring manipulation skills from humans to robots, 2026b. URL <https://arxiv.org/abs/2606.28133>.
- Yi Chen, Yuying Ge, Hui Zhou, Mingyu Ding, Yixiao Ge, and Xihui Liu. Dial: Decoupling intent and action via latent world modeling for end-to-end vla, 2026c. URL <https://arxiv.org/abs/2603.29844>.
- Ying Chen, Weizhen Li, Zhe Hu, Zhenjiang Li, Rui Jiang, Zhifeng Gu, Lihuang Fang, Jiangping Liu, Lei Yi, and Jie Chen. Capek 0.5: An execution-centric vision-language model for embodied intelligence, 2026d. URL <https://arxiv.org/abs/2608.06756>.
- Cheng Chi, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion, 2023. URL <https://arxiv.org/abs/2303.04137>.
- Dima Damen, Hazel Doughty, Giovanni Maria Farinella, Antonino Furnari, Evangelos Kazakos, Jian Ma, Davide Moltisanti, Jonathan Munro, Toby Perrett, Will Price, and Michael Wray. Rescaling egocentric vision, 2022. URL <https://arxiv.org/abs/2006.13256>.

- Senyu Fei, Xiaopeng Yu, Siyin Wang, Xianzhong Zhao, Jingjing Gong, and Xipeng Qiu. Wcm: A world critic model for vision-language-action reinforcement learning, 2026. URL <https://arxiv.org/abs/2607.29613>.
- Aaron Ferguson, Ahmed A. A. Osman, Berta Bescos, et al. MHR: Momentum human rig, 2025. URL <https://arxiv.org/abs/2511.15586>.
- Galaxea Team. Galaxea g0.5 technical report. <https://opengalaxea.github.io/G05/>, 2026.
- Ankit Goyal, Hugo Hadfield, Xuning Yang, Valts Blukis, and Fabio Ramos. Vla-0: Building state-of-the-art vlas with zero modification, 2025. URL <https://arxiv.org/abs/2510.13054>.
- Raghav Goyal, Samira Ebrahimi Kahou, Vincent Michalski, Joanna Materzyńska, Susanne Westphal, Heuna Kim, Valentin Haenel, Ingo Fruend, Peter Yianilos, Moritz Mueller-Freitag, Florian Hoppe, Christian Thureau, Ingo Bax, and Roland Memisevic. The "something something" video database for learning and evaluating visual common sense, 2017. URL <https://arxiv.org/abs/1706.04261>.
- Kristen Grauman, Andrew Westbury, Eugene Byrne, Zachary Chavis, Antonino Furnari, Rohit Girdhar, Jackson Hamburger, Hao Jiang, Miao Liu, Xingyu Liu, Miguel Martin, Tushar Nagarajan, Ilija Radosavovic, Santhosh Kumar Ramakrishnan, Fiona Ryan, Jayant Sharma, Michael Wray, Mengmeng Xu, Eric Zhongcong Xu, Chen Zhao, Siddhant Bansal, Dhruv Batra, Vincent Cartillier, Sean Crane, Tien Do, Morrie Doulaty, Akshay Erapalli, Christoph Feichtenhofer, Adriano Fragomeni, Qichen Fu, Abrahm Gebreselasie, Cristina Gonzalez, James Hillis, Xuhua Huang, Yifei Huang, Wenqi Jia, Weslie Khoo, Jachym Kolar, Satwik Kottur, Anurag Kumar, Federico Landini, Chao Li, Yanghao Li, Zhenqiang Li, Karttikeya Mangalam, Raghava Modhugu, Jonathan Munro, Tullie Murrell, Takumi Nishiyasu, Will Price, Paola Ruiz Puentes, Merey Ramazanov, Leda Sari, Kiran Somasundaram, Audrey Southerland, Yusuke Sugano, Ruijie Tao, Minh Vo, Yuchen Wang, Xindi Wu, Takuma Yagi, Ziwei Zhao, Yunyi Zhu, Pablo Arbelaez, David Crandall, Dima Damen, Giovanni Maria Farinella, Christian Fuegen, Bernard Ghanem, Vamsi Krishna Ithapu, C. V. Jawahar, Hanbyul Joo, Kris Kitani, Haizhou Li, Richard Newcombe, Aude Oliva, Hyun Soo Park, James M. Rehg, Yoichi Sato, Jianbo Shi, Mike Zheng Shou, Antonio Torralba, Lorenzo Torresani, Mingfei Yan, and Jitendra Malik. Ego4d: Around the world in 3,000 hours of egocentric video, 2022. URL <https://arxiv.org/abs/2110.07058>.
- Kristen Grauman, Andrew Westbury, Lorenzo Torresani, Kris Kitani, Jitendra Malik, Triantafyllos Afouras, Kumar Ashutosh, Vijay Baiyya, Siddhant Bansal, Bikram Boote, Eugene Byrne, Zach Chavis, Joya Chen, Feng Cheng, Fu-Jen Chu, Sean Crane, Avijit Dasgupta, Jing Dong, Maria Escobar, Cristhian Forigua, Abrahm Gebreselasie, Sanjay Haresh, Jing Huang, Md Mohaiminul Islam, Suyog Jain, Rawal Khirodkar, Devansh Kukreja, Kevin J Liang, Jia-Wei Liu, Sagnik Majumder, Yongsan Mao, Miguel Martin, Effrosyni Mavroudi, Tushar Nagarajan, Francesco Ragusa, Santhosh Kumar Ramakrishnan, Luigi Seminara, Arjun Somayazulu, Yale Song, Shan Su, Zihui Xue, Edward Zhang, Jinxu Zhang, Angela Castillo, Changan Chen, Xinzhu Fu, Ryosuke Furuta, Cristina Gonzalez, Prince Gupta, Jiabo Hu, Yifei Huang, Yiming Huang, Weslie Khoo, Anush Kumar, Robert Kuo, Sach Lakhavani, Miao Liu, Mi Luo, Zhengyi Luo, Brigid Meredith, Austin Miller, Oluwatumini Oguntola, Xiaqing Pan, Penny Peng, Shraman Pramanick, Merey Ramazanov, Fiona Ryan, Wei Shan, Kiran Somasundaram, Chenan Song, Audrey Southerland, Masatoshi Tateno, Huiyu Wang, Yuchen Wang, Takuma Yagi, Mingfei Yan, Xitong Yang, Zecheng Yu, Shengxin Cindy Zha, Chen Zhao, Ziwei Zhao, Zhifan Zhu, Jeff Zhuo, Pablo Arbelaez, Gedas Bertasius, David Crandall, Dima Damen, Jakob Engel, Giovanni Maria Farinella, Antonino Furnari, Bernard Ghanem, Judy Hoffman, C. V. Jawahar, Richard Newcombe, Hyun Soo Park, James M. Rehg, Yoichi Sato, Manolis Savva, Jianbo Shi, Mike Zheng Shou, and Michael Wray. Ego-exo4d: Understanding skilled human activity from first- and third-person perspectives, 2024. URL <https://arxiv.org/abs/2311.18259>.
- Lingxiao Guo, Huanyu Li, and Guanya Shi. Worlds in one demo: A synthetic data engine for learning open-world mobile manipulation, 2026. URL <https://arxiv.org/abs/2607.13154>.
- Joey Hejna, Chethan Bhateja, Yichen Jiang, Karl Pertsch, and Dorsa Sadigh. Re-mix: Optimizing data mixtures for large scale imitation learning, 2024. URL <https://arxiv.org/abs/2408.14037>.
- Ryan Hoque, Peide Huang, David J. Yoon, Mouli Sivapurapu, and Jian Zhang. Egodex: Learning dexterous manipulation from large-scale egocentric video, 2025. URL <https://arxiv.org/abs/2505.11709>.

- JiaKui Hu, Shanshan Zhao, Qing-Guo Chen, Xuerui Qiu, Jialun Liu, Zhao Xu, Weihua Luo, Kaifu Zhang, and Yanye Lu. Omni-view: Unlocking how generation facilitates understanding in unified 3d model based on multiview images, 2025. URL <https://arxiv.org/abs/2511.07222>.
- Yucheng Hu, Jianke Zhang, Yuanfei Luo, Yanjiang Guo, Xiaoyu Chen, Xinshu Sun, Kun Feng, Qingzhou Lu, Sheng Chen, Yangang Zhang, Wei Li, and Jianyu Chen. Bagelvla: Enhancing long-horizon manipulation via interleaved vision-language-action generation, 2026. URL <https://arxiv.org/abs/2602.09849>.
- Wenhui Huang, Changhe Chen, Han Qi, Chen Lv, Yilun Du, and Heng Yang. Motvla: A vision-language-action model with unified fast-slow reasoning, 2025. URL <https://arxiv.org/abs/2510.18337>.
- Wenhui Huang, Songyan Zhang, Qihang Huang, Zhidong Wang, Zhiqi Mao, Collister Chua, Zhan Chen, Long Chen, and Chen Lv. Automot: A unified vision-language-action model with asynchronous mixture-of-transformers for end-to-end autonomous driving, 2026. URL <https://arxiv.org/abs/2603.14851>.
- Tao Jiang, Tianyuan Yuan, Yicheng Liu, Chenhao Lu, Jianning Cui, Xiao Liu, Shuiqi Cheng, Jiyang Gao, Huazhe Xu, and Hang Zhao. Galaxea open-world dataset and g0 dual-system vla model, 2025. URL <https://arxiv.org/abs/2509.00576>.
- Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models, 2020. URL <https://arxiv.org/abs/2001.08361>.
- Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. Openvla: An open-source vision-language-action model, 2024. URL <https://arxiv.org/abs/2406.09246>.
- Filippo Lazzati, Kyle Stachowicz, William Chen, Alberto Maria Metelli, Andrew Wagenmaker, and Sergey Levine. Why does action chunking improve behavioral cloning performance in robotic control?, 2026. URL <https://arxiv.org/abs/2608.02547>.
- Baoyu Li, Xinchun Yin, Mengying Lin, Yixin Zhang, and Danfei Xu. Egowam: World action models beyond pixels with in-the-wild egocentric human data, 2026a. URL <https://arxiv.org/abs/2607.08436>.
- Junfeng Li, Junjie He, Zhide Zhong, Yangyang Zheng, Pingyue Sheng, Jiayu Dong, Ruixin Li, Haodong Yan, Jiaguan Zhu, Tianran Zhang, Runze Yu, Wen Chen, Liuqing Yang, Yuxiang Gao, and Haoang Li. Dypes-vla: Learning shared dynamics priors and embodiment-specific control for cross-embodiment manipulation, 2026b. URL <https://arxiv.org/abs/2608.06374>.
- Lin Li, Qihang Zhang, Yiming Luo, Shuai Yang, Ruilin Wang, Fei Han, Mingrui Yu, Zelin Gao, Nan Xue, Xing Zhu, Yujun Shen, and Yinghao Xu. Causal world modeling for robot control, 2026c. URL <https://arxiv.org/abs/2601.21998>.
- Qixiu Li, Yu Deng, Yaobo Liang, Lin Luo, Lei Zhou, Chengtang Yao, Lingqi Zeng, Zhiyuan Feng, Huizhi Liang, Sicheng Xu, Yizhong Zhang, Xi Chen, Hao Chen, Lily Sun, Dong Chen, Jiaolong Yang, and Baining Guo. Scalable vision-language-action model pretraining for robotic manipulation with real-life human activity videos, 2025a. URL <https://arxiv.org/abs/2510.21571>.
- Runhao Li, Wenkai Guo, Zhenyu Wu, Changyuan Wang, Haoyuan Deng, Zhenyu Weng, Yap-Peng Tan, and Ziwei Wang. Map-vla: Memory-augmented prompting for vision-language-action model in robotic manipulation, 2025b. URL <https://arxiv.org/abs/2511.09516>.
- Zhe Li, Zhenzhe Zhang, Yangyang Wei, Wenjie Zhang, Xichen Yuan, Peiyuan Zhi, Gen Li, Xinying Guo, Fengjie Gao, Jianfei Yang, and Shanghang Zhang. ω -0: A latent predictive world action model for concurrent humanoid loco-manipulation, 2026d. URL <https://arxiv.org/abs/2608.06375>.

- Weixin Liang, Lili Yu, Liang Luo, Srinivasan Iyer, Ning Dong, Chunting Zhou, Gargi Ghosh, Mike Lewis, Wen tau Yih, Luke Zettlemoyer, and Xi Victoria Lin. Mixture-of-transformers: A sparse and scalable architecture for multi-modal foundation models, 2025. URL <https://arxiv.org/abs/2411.04996>.
- Fanqi Lin, Yingdong Hu, Pingyue Sheng, Chuan Wen, Jiacheng You, and Yang Gao. Data scaling laws in imitation learning for robotic manipulation, 2025a. URL <https://arxiv.org/abs/2410.18647>.
- Fanqi Lin, Ruiqian Nai, Yingdong Hu, Jiacheng You, Junming Zhao, and Yang Gao. Onetwovla: A unified vision-language-action model with adaptive reasoning, 2025b. URL <https://arxiv.org/abs/2505.11917>.
- Fanqi Lin, Kushal Arora, Jean Mercat, Haruki Nishimura, Paarth Shah, Chen Xu, Mengchao Zhang, Mark Zolotas, Maya Angeles, Owen Pfannenstiehl, Andrew Beaulieu, and Jose Barreiros. A systematic study of data modalities and strategies for co-training large behavior models for robot manipulation, 2026a. URL <https://arxiv.org/abs/2602.01067>.
- Haotong Lin, Sili Chen, Jun Hao Liew, Donny Y. Chen, Zhenyu Li, Guang Shi, Jiashi Feng, and Bingyi Kang. Depth anything 3: Recovering the visual space from any views. *arXiv preprint arXiv:2511.10647*, 2025c.
- Nie Lin, Takehiko Ohkawa, Sijin Chen, Ruoshi Wen, Zhuohang Li, Liqun Huang, Zhengming Zhu, Yiming Bao, Yunfei Li, Minjie Cai, Xiao Ma, Wei Xu, and Yoichi Sato. Simdex: Mining similar egocentric videos for cross-embodiment dexterous manipulation, 2026b. URL <https://arxiv.org/abs/2608.04196>.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling, 2023. URL <https://arxiv.org/abs/2210.02747>.
- Peidong Liu, Yongce Liu, Songyan Guo, Fuyuan Ma, Zhihao Yuan, Ao Li, Zengjue Chen, Wenhao Li, Tianle Zhang, Mingyang Li, Jiale Zhang, Junzhe Xiong, Zhiyuan Xiang, Dafeng Chi, Yuzheng Zhuang, Liyi Luo, Wei Tan, Dongjiang Li, Nan Jiang, Yihang Li, Qingrong He, Jiaming Liang, Chen Cai, Mingxi Luo, Hui Zhang, Peng Hao, Song Wang, Ning Qiao, Yince Gao, Lei Kang, Junwu Xiong, Ruodai Li, Jiawei Li, Hui Shen, Yicheng Gong, Nan Duan, and Liang Lin. Joyai-sim: A simulation-enabled interconversion toolchain for the embodied data pyramid, 2026. URL <https://arxiv.org/abs/2606.16776>.
- Songming Liu, Lingxuan Wu, Bangguo Li, Hengkai Tan, Huayu Chen, Zhengyi Wang, Ke Xu, Hang Su, and Jun Zhu. Rdt-1b: a diffusion foundation model for bimanual manipulation, 2025. URL <https://arxiv.org/abs/2410.07864>.
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. URL <https://arxiv.org/abs/1711.05101>.
- Hao Luo, Wanpeng Zhang, Yicheng Feng, Sipeng Zheng, Haiweng Xu, Chaoyi Xu, Ziheng Xi, Yuhui Fu, and Zongqing Lu. Being-h0.7: A latent world-action model from egocentric videos, 2026. URL <https://arxiv.org/abs/2605.00078>.
- Haolang Ma, Junhao Cai, Xiaoxu Xu, Hao Li, Yuyin Yang, Yang Tian, Jiafei Cao, Hongrui Zhu, Zherui Qiu, Zhaxizhuoma, Yuqiang Yang, Jiaqi Peng, Xueyuan Wei, Yangkun Zhu, Jiahao Jiang, Xing Gao, Hanqing Wang, Feng Yuan, Kailin Li, Xueyue Zhu, Tai Wang, Yan Ding, Jiangmiao Pang, Jia Zeng, Jingjing Zhang, Bowen Zhou, Yao Mu, Chunhua Shen, and Weinan Zhang. Internvla-a1.5: Unifying understanding, latent foresight, and action for compositional generalization, 2026a. URL <https://arxiv.org/abs/2607.04988>.
- Teli Ma, Jia Zheng, Zifan Wang, Chunli Jiang, Andy Cui, Junwei Liang, and Shuo Yang. Dit4dit: Jointly modeling video dynamics and actions for generalizable robot control, 2026b. URL <https://arxiv.org/abs/2603.10448>.
- NVIDIA. Alpamayo-r1: Bridging reasoning and action prediction for generalizable autonomous driving in the long tail, 2025a. URL <https://arxiv.org/abs/2511.00088>.

- NVIDIA. Gr00t n1: An open foundation model for generalist humanoid robots, 2025b. URL <https://arxiv.org/abs/2503.14734>.
- NVIDIA. Cosmos 3: Omnimodal world models for physical ai, 2026. URL <https://arxiv.org/abs/2606.02800>.
- Octo Model Team. Octo: An open-source generalist robot policy, 2024. URL <https://arxiv.org/abs/2405.12213>.
- Open X-Embodiment Collaboration. Open x-embodiment: Robotic learning datasets and rt-x models, 2024. URL <https://arxiv.org/abs/2310.08864>.
- William Peebles and Saining Xie. Scalable diffusion models with transformers, 2023. URL <https://arxiv.org/abs/2212.09748>.
- Karl Pertsch, Kyle Stachowicz, Brian Ichter, Danny Driess, Suraj Nair, Quan Vuong, Oier Mees, Chelsea Finn, and Sergey Levine. Fast: Efficient action tokenization for vision-language-action models, 2025. URL <https://arxiv.org/abs/2501.09747>.
- Physical Intelligence. $\pi_{0.5}$: a vision-language-action model with open-world generalization, 2025a. URL <https://arxiv.org/abs/2504.16054>.
- Physical Intelligence. $\pi_{0.6}^*$: a vla that learns from experience, 2025b. URL <https://arxiv.org/abs/2511.14759>.
- Physical Intelligence. $\pi_{0.7}$: a steerable generalist robotic foundation model with emergent capabilities, 2026. URL <https://arxiv.org/abs/2604.15483>.
- Lu Qiu, Yizhuo Li, Yi Chen, Yuying Ge, Yixiao Ge, and Xihui Liu. Making foresight actionable: Repurposing representation alignment in world action models, 2026. URL <https://arxiv.org/abs/2606.12217>.
- Qwen Team. Qwen3.5: Towards native multimodal agents, February 2026a. URL <https://qwen.ai/blog?id=qwen3.5>. Model card: <https://huggingface.co/Qwen/Qwen3.5-2B>.
- Qwen Team. Qwen3.6-35B-A3B: Agentic coding power, now open to all. <https://qwen.ai/blog?id=qwen3.6-35b-a3b>, April 2026b. URL <https://qwen.ai/blog?id=qwen3.6-35b-a3b>.
- Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. Dinov3, 2025. URL <https://arxiv.org/abs/2508.10104>.
- Jingwen Sun, Wen Yao Zhang, Zekun Qi, Shaojie Ren, Zezhi Liu, Hanxin Zhu, Guangzhong Sun, Xin Jin, and Zhibo Chen. Vla-jepa: Enhancing vision-language-action model with latent world model, 2026. URL <https://arxiv.org/abs/2602.10098>.
- RA Team. Joyai-ra 0.5: Scaling robot manipulation learning via dual action alignment, 2026. URL <https://arxiv.org/abs/2608.05674>.
- Team Wan. Wan: Open and advanced large-scale video generative models, 2025. URL <https://arxiv.org/abs/2503.20314>.
- Marcel Torne, Karl Pertsch, Homer Walke, Kyle Vedder, Suraj Nair, Brian Ichter, Allen Z. Ren, Haohuan Wang, Jiaming Tang, Kyle Stachowicz, Karan Dhabalia, Michael Equi, Quan Vuong, Jost Tobias Springenberg, Sergey Levine, Chelsea Finn, and Danny Driess. Mem: Multi-scale embodied memory for vision language action models, 2026. URL <https://arxiv.org/abs/2603.03596>.

- Qiuyue Wang, Mingsheng Li, Jian Guan, Jinhui Ye, Sicheng Xie, Yitao Liu, Junhao Chen, Zhixuan Liang, Jie Zhang, Xintong Hu, Xuhong Huang, Pei Lin, Junyang Lin, Dayiheng Liu, Shuai Bai, Jingren Zhou, Jiazhaoh Zhang, Haoqi Yuan, Gengze Zhou, Hang Yin, Ye Wang, Yiyang Huang, Zixing Lei, Wujian Peng, Delin Chen, Yingming Zheng, Jingyang Fan, Xianwei Zhuang, Xin Zhou, Haoyang Li, Anzhe Chen, Tong Zhang, Xuejing Liu, Yuchong Sun, Ruizhe Chen, Zhaozhai Li, Chenxu Lü, Zhibo Yang, Tao Yu, and Xionghui Chen. Qwen-vla: Unifying vision-language-action modeling across tasks, environments, and robot embodiments, 2026a. URL <https://arxiv.org/abs/2605.30280>.
- Ye Wang, Pei Lin, Xiong-Hui Chen, Haoqi Yuan, Zhixuan Liang, Yiyang Huang, Anzhe Chen, Zixing Lei, Jie Zhang, Tao Zhang, Haoyang Li, Tong Zhang, Chenxi Xiao, Ziyuan Jiao, and Qin Jin. Ego2robot: Scalable robot data synthesis from egocentric human data, 2026b. URL <https://arxiv.org/abs/2608.02580>.
- Yuqi Wang, Xinghang Li, Wenxuan Wang, Junbo Zhang, Yingyan Li, Yuntao Chen, Xinlong Wang, and Zhaoxiang Zhang. Unified vision-language-action model, 2025. URL <https://arxiv.org/abs/2506.19850>.
- Jiacheng Wei, Jerry Bai, Xiaoyu Yue, Zidong Wang, Xiaoyang Guo, Cheng Chen, Fanqi Pu, Fan Wu, Zhixu Yue, Yizhuo Li, Feng Qiu, Bo Liu, Yuying Ge, Hui Zhou, Chenyi Chen, and Yixiao Ge. XPACE: Joint world and action modeling from heterogeneous experience. Technical report, XPENG Robotics, 2026. URL <https://xpeng-robotics.github.io/xpace/>.
- Wei Wu, Fan Lu, Yunnan Wang, Shuai Yang, Shi Liu, Fangjing Wang, Qian Zhu, He Sun, Yong Wang, Shuailei Ma, Yiyu Ren, Kejia Zhang, Hui Yu, Jingmei Zhao, Shuai Zhou, Zhenqi Qiu, Houlong Xiong, Ziyu Wang, Zechen Wang, Ran Cheng, Yong-Lu Li, Yongtao Huang, Xing Zhu, Yujun Shen, and Kecheng Zheng. A pragmatic vla foundation model, 2026a. URL <https://arxiv.org/abs/2601.18692>.
- Xianjin Wu, Dingkan Liang, Tianrui Feng, Kui Xia, Yumeng Zhang, Xiaofan Li, Xiao Tan, and Xiang Bai. Generation models know space: Unleashing implicit 3d priors for scene understanding, 2026b. URL <https://arxiv.org/abs/2603.19235>.
- Wei Xiao, Weiliang Tang, Yuying Ge, Hui Zhou, Yao Mu, Li Zhang, and Yixiao Ge. Rove: Unlocking human interventions for humanoid manipulation via reinforcement learning, 2026. URL <https://arxiv.org/abs/2606.17011>.
- XPeng Robotics. VIGIL: Terminal commitment in embodied evaluation. <https://xpeng-robotics.github.io/vigil/>, 2026.
- Charles Xu, Jost Tobias Springenberg, Michael Equi, Ali Amin, Adnan Esmail, Sergey Levine, and Liyiming Ke. RL token: Bootstrapping online rl with vision-language-action models, 2026. URL <https://arxiv.org/abs/2604.23073>.
- Seonghyeon Ye, Joel Jang, Byeongguk Jeon, Sejeun Joo, Jianwei Yang, Baolin Peng, Ajay Mandlekar, Reuben Tan, Yu-Wei Chao, Bill Yuchen Lin, Lars Liden, Kimin Lee, Jianfeng Gao, Luke Zettlemoyer, Dieter Fox, and Minjoon Seo. Latent action pretraining from videos, 2025. URL <https://arxiv.org/abs/2410.11758>.
- Seonghyeon Ye, Yunhao Ge, Kaiyuan Zheng, Shenyuan Gao, Sihyun Yu, George Kurian, Suneel Indupuru, You Liang Tan, Chuning Zhu, Jiannan Xiang, Ayaan Malik, Kyungmin Lee, William Liang, Nadun Ranawaka, Jiasheng Gu, Yinzhen Xu, Guanzhi Wang, Fengyuan Hu, Avnish Narayan, Johan Bjorck, Jing Wang, Gwanghyun Kim, Dantong Niu, Ruijie Zheng, Yuqi Xie, Jimmy Wu, Qi Wang, Ryan Julian, Danfei Xu, Yilun Du, Yevgen Chebotar, Scott Reed, Jan Kautz, Yuke Zhu, Linxi "Jim" Fan, and Joel Jang. World action models are zero-shot policies, 2026. URL <https://arxiv.org/abs/2602.15922>.
- Haoqi Yuan, Zhixuan Liang, Anzhe Chen, Ye Wang, Haoyang Li, Pei Lin, Yiyang Huang, Zixing Lei, Tong Zhang, Jiazhaoh Zhang, Jie Zhang, Jingyang Fan, Gengze Zhou, Qihang Peng, Chenxu Lv, Xiaoyue Chen, An Yang, Fei Huang, Junyang Lin, Dayiheng Liu, Jingren Zhou, Chenfei Wu, and Xiong-Hui Chen. Qwen-robotmanip technical report: Alignment unlocks scale for robotic manipulation foundation models, 2026a. URL <https://arxiv.org/abs/2606.17846>.

- Tianyuan Yuan, Zibin Dong, Yicheng Liu, and Hang Zhao. Fast-wam: Do world action models need test-time future imagination?, 2026b. URL <https://arxiv.org/abs/2603.16666>.
- Lihan Zha, Asher J. Hancock, Mingtong Zhang, Tenny Yin, Yixuan Huang, Dhruv Shah, Allen Z. Ren, and Anirudha Majumdar. Lap: Language-action pre-training enables zero-shot cross-embodiment transfer, 2026. URL <https://arxiv.org/abs/2602.10556>.
- He Zhang, Lingzhu Xiang, Haitao Lin, Zeyu Huang, Minghui Wang, Dingyan Zhong, Yubo Dong, Yihao Wu, Yongming Rao, Dongsheng Zhang, Wanjia He, Ling Chen, Kai Huang, Jiahao Chen, Sichang Su, Xumin Yu, Ziyi Wang, Chengwei Zhu, Xiao Teng, Yuchun Guo, Yufeng Zhang, Yuandong Liu, Rui Wang, Zisheng Lu, Han Hu, and Zhengyou Zhang. Hy-embodied-0.5-vla: From vision-language-action models to a real-world robot learning stack, 2026. URL <https://arxiv.org/abs/2606.14409>.
- Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bimanual manipulation with low-cost hardware, 2023. URL <https://arxiv.org/abs/2304.13705>.
- Ruijie Zheng, Dantong Niu, Yuqi Xie, Jing Wang, Mengda Xu, Yunfan Jiang, Fernando Castañeda, Fengyuan Hu, You Liang Tan, Letian Fu, Trevor Darrell, Furong Huang, Yuke Zhu, Danfei Xu, and Linxi Fan. Egoscale: Scaling dexterous manipulation with diverse egocentric human data, 2026. URL <https://arxiv.org/abs/2602.16710>.
- Xiang Zhu, Puzhen Yuan, Yichen Liu, and Jianyu Chen. Harp-vla: Human-robot aligned representation learning for vision-language-action model, 2026. URL <https://arxiv.org/abs/2605.31234>.

Contributors

Huimin Pan^{1*}, Yufan Ren^{1*†}, Kunpeng Song^{1*}, Siyang Wang^{1*}, Xiwen Zhang^{1*}, Xiaoyun Hu^{1,2§}, Zhuoxu Duan^{1,3§}, Hanrui Zheng^{1,4§}, Jialeng Ni^{1,5§}, Nathan Zhao¹, Sibom Ma^{1,6§}, Zhenxuan Fan^{1,7§}, Zhongyang Che¹, Danny Bao¹, Jiacheng Wei¹, Jerry Bai¹, Xiaoyu Yue¹, Xiaoyang Guo¹, Chenyi Chen^{1‡}

* Equal contribution, listed alphabetically by last name. † Project lead. ‡ Supervision. § Work done during an internship at XPENG Robotics.

¹ XPENG Robotics ² Tongji University ³ Rensselaer Polytechnic Institute ⁴ University of Science and Technology of China ⁵ University of Michigan ⁶ University of British Columbia ⁷ Zhejiang University

Acknowledgements

We thank members of the XPENG Robotics team—Zike Cheng, Ruixuan Zhao, Mengen Luo, Luoyu Bai, Ying Zhang, Haoyue Liu, Peipeng Chen, Ruyi Li, Yizhe Liu, Yongqi Meng, Yinggan Xu, Xi Chen, Bin Shen, and Yuying Ge—for insightful discussions and deployment support, and Zhenglong Du, Hongmei Gan, and Weiqi Liang for assistance with teleoperation and data collection.